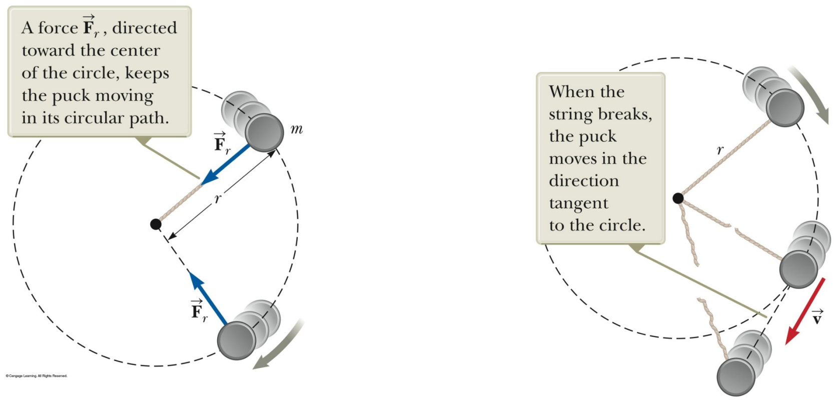
Chapter 6: Circular motion and other forces

Centripetal force
Resistive forces

Uniform Circular Motion



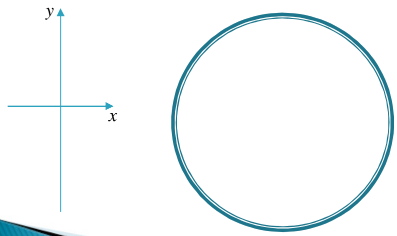
Newton's second law – applied along the radial direction

$$\sum F = ma_c = m\frac{v^2}{r}$$

Quizz

You are riding on a Ferris wheel that is rotating with constant speed. The car in which you are riding always maintains an upright orientation (it does not invert).

- Determine the normal force on you from the seat when you are at the top of the ride? Give the direction and a general equation for the normal force.
- What is the direction of the net force on you when you are at the top of the ride?

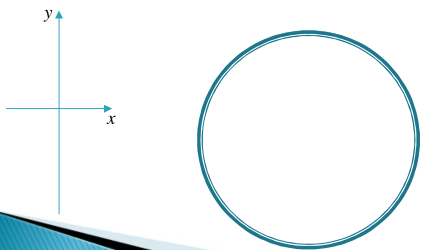


See Example 6.5 (p 132)

Quizz

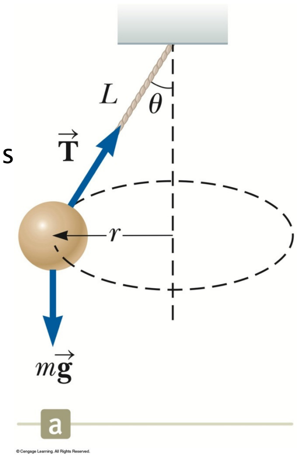
If the ride is no longer one of those where you are always upright, but you are strapped to your seat with a safety harness. The cage in which you are seated is still rotating with constant speed, but when you are at the top of the ride you are actually upside down.

- What is the direction of the normal force on you from the seat when you are at the top of the ride?
- What is the direction of the net force on you when you are at the top of the ride?



Example 6.1

A small ball of mass m is suspended from a string of length L. The ball revolves with constant speed v in a horizontal circle of radius r as shown in the figure (Conical pendulum). Find an expression for v in terms of L and θ.



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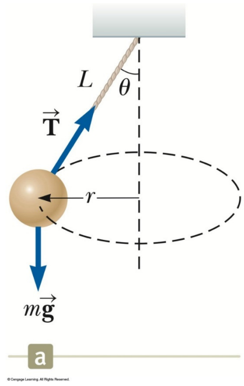
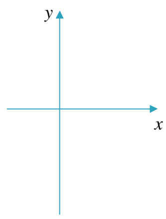


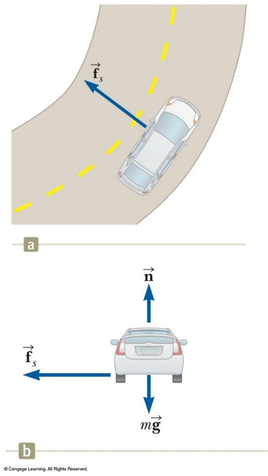
1. Identify the forces acting on the ball.
2. Draw free body diagram of the ball.
3. Apply Newton's second law in x– and y–directions.

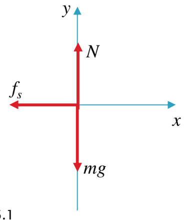


Example 6.3

A 1500 kg car moving on a flat, horizontal road negotiates a curve as shown in the figure. If the radius of the curve is 35.0 m and the coefficient of static friction between the tires and the dry road is 0.523, find the maximum speed the car can have to still make the turn successfully.

1. Identify the forces acting on the car.
2. Draw free body diagram of the car.
3. Apply Newton's second law in *x*– and *y*– directions.





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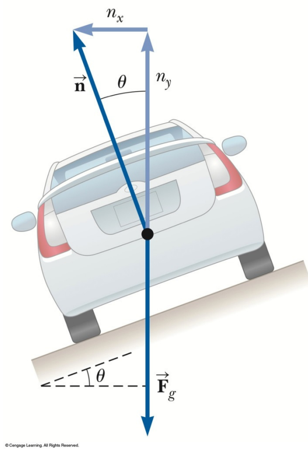
Example 6.4

A civil engineer wishes to redesign the curved roadway in such a way that a car will not have to rely on friction to round the curve without skidding. A banked roadway is used, which is tilted towards the inside of the curve, as seen in the figure. If the designated speed for the road is13.4 m/s (48 km/h) and the radius of the curve is 35.0 m, at what angle should the curve be banked?



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Example 6.4

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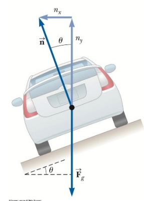

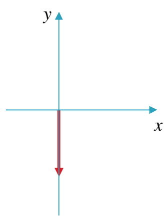

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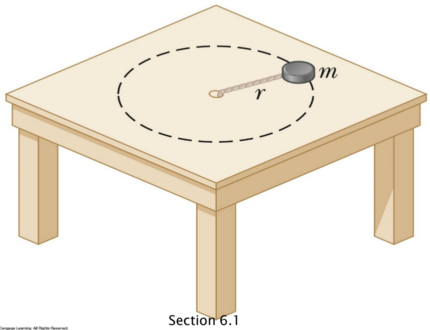
Problem

A light string can support a stationary hanging load of 25 kg before breaking. An object of mass 3.00 kg attached to the string rotates on a frictionless, horizontal table in a circle of radius 0.800 m, and the other end of the string is held fixed. What is the maximum speeds the object can have without the string breaking?



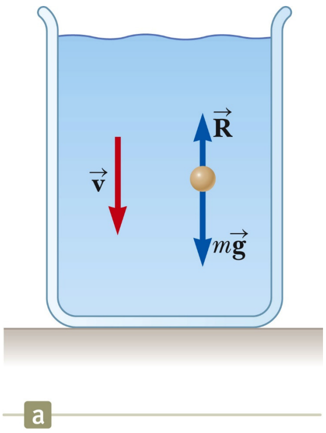
Motion in the presence of Resistive Forces

- Motion can be through a medium.
  - Either a liquid or a gas
- The medium exerts a *resistive force*, $\vec{\mathbf{R}}$, on an object moving through the medium.
- The magnitude of $\vec{\mathbf{R}}$ depends on the medium.
- The direction of $\vec{\mathbf{R}}$ is opposite the direction of motion of the object relative to the medium.
  - This direction may or may not be in the direction opposite the object's velocity according to the observer .
- $\vec{\mathbf{R}}$ nearly always increases with increasing speed.

Resistive Force Proportional To Speed

- The resistive force can be expressed as
  $\vec{\mathbf{R}} = -b\vec{\mathbf{v}}$
- b depends on the property of the medium, and on the shape and dimensions of the object.
- The negative sign indicates $\vec{\mathbf{R}}$ is in the opposite direction to $\vec{\mathbf{v}}$.

Resistive Force Proportional To Speed



$$\sum F_y = -ma = bv - mg$$

$$m\frac{dv}{dt} = mg - bv$$

$$\frac{dv}{dt} = g - \frac{b}{v}v$$

Resistive Force Proportional To v^2

- For objects moving at high speeds through air, the resistive force is modelled as a function of the square of the speed.
- $R = \frac{1}{2} D\rho A v^2$
  - D is a dimensionless empirical determined quantity called the drag coefficient.
  - ρ is the density of air.
  - A is the cross-sectional area of the object.
  - v is the speed of the object.

Resistive Force Proportional To v^2

- Analysis of an object falling through air accounting for air resistance.

$$\sum F = -ma = \frac{1}{2} D\rho A v^2 - mg$$

$$a = g - \left(\frac{D\rho A}{2m}\right) v^2$$

- The terminal speed will occur when the acceleration goes to zero.
- From the previous equation

$$v_T = \sqrt{\frac{2mg}{D\rho A}}$$